MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER TYPE 2N2084

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a PNP, germanium, high-frequency, low-power transistor.

1.2 Physical dimensions. See figure 1 (TO-33).

* 1.3 Maximum ratings.

P_T 1/	V_{CBO}	V_{CEO}	V_{CER} R_{BE} = 2 kilohms	V_{EBO}	I_C	T_{stg} and T_{op}
mW	Vdc	Vdc	Vdc	Vdc	mAdc	°C
125	-40	-20	-40	-1	-10	-65 to +100

1/ Derate linearly 1.67 mW/°C for $T_A > 25$°C.

* 1.4 Primary electrical characteristics.

Limits	h_{fe} V_{CE} = -6 Vdc I_C = -1 mAdc	h_{FE} V_{CE} = -6 Vdc I_C = -1 mAdc	NF f = 100 MHz	G_{pb} f = 100 MHz	f_t V_{CE} = -6 Vdc I_C = -1 mAdc f = 35 MHz
			dB	dB	MHz
Min	40	40	---	12.5	40
Max	250	250	11	---	160

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

* MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

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* 3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows:

Y_{ie} . . . . . Small-signal short-circuit input admittance (common emitter).
C_{re} . . . . . Small-signal short-circuit reverse capacitance.
$|Z_{rb}|$ . . . . Absolute value of small-signal, open-circuit reverse impedance transfer ratio.

3.3 Design, construction, and physical dimensions. Transistor shall be of the design, construction, and physical dimensions shown on figure 1.

* 3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold-plated. (Leads may be tin-coated if specified in the contract or order, and this requirement shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking, see 6.2.)

* 3.3.1.1 Selectivity of lead material. Where choice of lead material (see 3.3.1) is desired, it shall be specified in the contract or order (see 6.2).

* 3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

* 4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

* 4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

* 4.3.3 Group C inspection. Group C inspection shall consist of the tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

* 4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hours life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria, see 4.3.3.

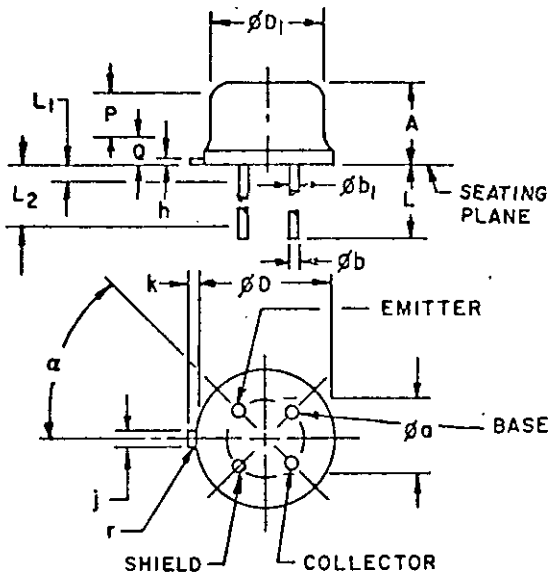
* 4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

* 4.4.1 Condition of the shield lead. The shield lead is open for the following tests: BV_{CBO}, BV_{EBO}, BV_{CEO}, BV_{CER}, I_{CBO}, C_{ibo}, and C_{obo}. The shield lead is grounded for the following tests: V_{BE}, h_{FE}, G_{pb}, f_t, h_{fe}, NF, $|Z_{rb}|$, C_{re}, and RE_{hie}.

* 4.4.2 Resistance to solvents. Transistors shall be subjected to tests in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the transistor body where marking has been applied shall be brushed.
(b) After subjection to the tests there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

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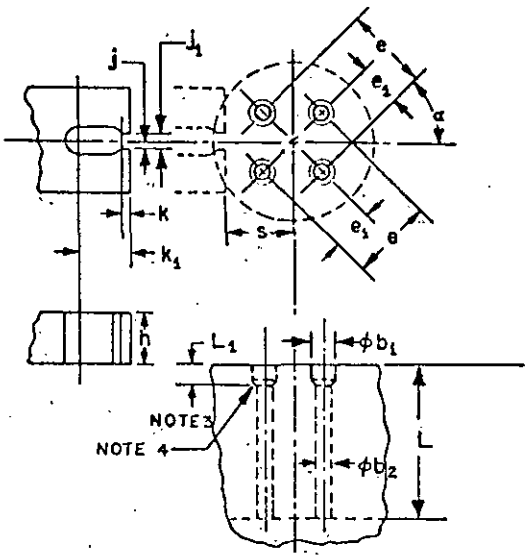


SYMBOL	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.240	.260	6.10	6.60	
Øa	.200 TP		5.080 TP		6
Øb	.016	.021	.407	.533	7, 8
$Øb_1$	.016	.019	.407	.482	7, 8
ØD	.335	.370	8.51	9.40	
$ØD_1$	.305	.335	7.75	8.51	
h	.009	.041	.23	1.04	
j	.028	.034	.712	.863	2
k	.029	.045	.74	1.14	3
L	1.500	1.750	38.10	44.45	7, 8
L_1	---	.050	---	1.27	7, 8
L_2	.250	---	6.35	---	7, 8
P	.100	---	2.54	---	5
Q	---	.050	---	1.27	4
r	---	.010	---	.254	10
α	45° TP		45° TP		6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Beyond r(radius) maximum, j shall be held for a minimum length of .011(.28 mm).
3. k measured from maximum ØD.
4. Outline in this zone is not controlled.
5. $ØD_1$ shall not vary more than .010(.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gage plane .054 + .001 - .000 (1.37 + .03 - .00 mm) below seating plane shall be within .007(.18 mm) radius of True Position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gage and gaging procedure shown in figure 2.
7. $Øb_1$ applies between L_1 and L_2. Øb applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
8. All four leads.
9. The shield lead shall be internally connected to the case. The other three leads shall be electrically insulated from the case.
10. r (radius) applies to both inside corners of tab.

FIGURE 1. Physical dimensions of transistor type 2N2084 (TO-33).

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SYMBOL	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
ϕb_1	.0595	.0605	1.511	1.537	
ϕb_2	.0325	.0335	.8255	.8509	
e	.1995	.2005	5.067	5.093	
e_1	.0995	.1005	2.527	2.552	
h	.150 NOM		3.81 NOM		
j	.0175	.0180	.4445	.4572	
j_1	.0350	.0355	.8890	.9017	
k	.009	.011	.23	.28	
k_1	.125 NOM		3.18 NOM		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
S	.182	.199	4.62	5.05	1
α	44.90°	45.10°	44.90°	45.10°	

NOTES:
1. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
2. The following gaging procedure shall be used:

   The device being measured shall be inserted until its seating plane is .125(3.18 mm) ± .010(.25 mm) from the seating surface of the gage. A force of 8 ± .5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.

   The use of a pin straightener prior to insertion in the gage is permissible.

3. Gaging plane.
4. Drill angle.

FIGURE 2. Gage for lead and tab location for transistor type 2N2084.

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TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
* Subgroup 2			5				
Breakdown voltage, collector to base	3001	Bias cond. D; I_C = -50 μAdc		BV_{CBO}	-40	---	Vdc
Breakdown voltage, collector to emitter	3011	Bias cond. D; I_C = -1 mAdc		BV_{CEO}	-20	---	Vdc
Breakdown voltage, collector to emitter	3011	Bias cond. B; I_C = -50 μAdc; R_{BE} = 2 kilohms		BV_{CER}	-40	---	Vdc
Breakdown voltage, emitter to base	3026	Bias cond. D; I_E = -50 μAdc		BV_{EBO}	-1	---	Vdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -6 Vdc		I_{CBO}	---	-8	μAdc
Base emitter voltage (nonsaturated)	3066	Test cond. B; V_{CE} = -6 Vdc; I_C = -1 mAdc		V_{BE}	-210	-330	mVdc
Forward-current transfer ratio	3076	V_{CE} = -6 Vdc; I_C = -1 mAdc		h_{FE}	40	250	---
Small-signal power gain	3256	V_{CB} = -6 Vdc; I_E = 1 mAdc; f = 100 MHz (see figure 3)		G_{pb}	12.5	---	dB
Small-signal power gain	3256	V_{CB} = -4 Vdc; I_E = 1.5 mAdc; f = 100 MHz (see figure 3)		G_{pb}	10	---	dB
* Subgroup 3			5				
Extrapolated unity gain frequency	3261	V_{CE} = -6 Vdc; I_C = -1 mAdc; f = 35 MHz		f_t	40	160	MHz
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -6 Vdc; I_C = -1 mAdc		h_{fe}	40	250	---
Open circuit output capacitance	3236	V_{CB} = -6 Vdc; I_E = 0; 100 kHz $\leq$ f $\leq$ 1 MHz		C_{obo}	---	4	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 Vdc; I_E = 0; 100 kHz $\leq$ f $\leq$ 1 MHz		C_{ibo}	---	30	pF

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TABLE I. Group A inspection - Continued.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 3 - continued							
Noise figure	3246	V_{CB} = -6 Vdc; I_E = 1 mAdc; f = 10.7 MHz; R_g = 150 ohms; BW = 2 MHz		NF	---	8	dB
Noise figure	3246	V_{CB} = -6 Vdc; I_E = 1 mAdc; f = 100 MHz; R_g = 68 ohms; BW = 5 MHz		NF	---	11	dB
Magnitude of small-signal, open-circuit reverse impedance transfer ratio	---	V_{CB} = -6 Vdc; I_E = 1 mAdc; f = 2 MHz (see figure 4)		$\|Z_{rb}\|$	---	100	ohms
Small-signal short-circuit reverse capacitance	---	V_{CE} = -6 Vdc; I_E = 1 mAdc; f = 0.45 MHz (see figure 5)		C_{re}	---	2.4	pF
Real part of small-signal short-circuit input impedance	3266	V_{CE} = -6 Vdc; I_E = -1 mAdc; f = 50 MHz		RE_{hie}	20	100	ohms
* Subgroup 4			10				
High-temperature operation:		T_A = +75°C					
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -6 Vdc		I_{CBO}	---	180	μAdc
Low-temperature operation:		T_A = -65°C					
Forward-current transfer ratio	3076	V_{CE} = -6 Vdc; I_C = -1 mAdc		h_{FE}	10	---	---

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	See figure 1		---	---	---	---
* Subgroup 2			15				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. A, except in step 3, T_A = +100° +5°, -0°C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A		---	---	---	---

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TABLE II. Group B inspection - continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2 - continued							
Hermetic seal	1071	Test cond. G or H for fine leaks; test cond. A, C, D, or F for gross leaks		---	---	$1x10^{-7}$	atm cc/s
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -6 Vdc		I_{CBO}	---	-8	μAdc
Forward-current transfer ratio	3076	V_{CE} = -6 Vdc; I_C = -1 mAdc		h_{FE}	40	250	---
* Subgroup 3			10				
Shock	2016	Nonoperating; 1,500 G; 0.5 ms; 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration, variable frequency	2056	Nonoperating		---	---	---	---
Constant acceleration	2006	20,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
* Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
End points:							
Hermetic seal	1071	Test cond. G or H for fine leaks; test cond. A, C, D, or F for gross leaks		---	---	$1x10^{-7}$	atm cc/s
* Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
* Subgroup 6			7				
High-temperature life (nonoperating)	1032	T_{stg} = +100°C; time = 340 hours (see 4.3.4)		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -6 Vdc		I_{CBO}	---	-16	μAdc

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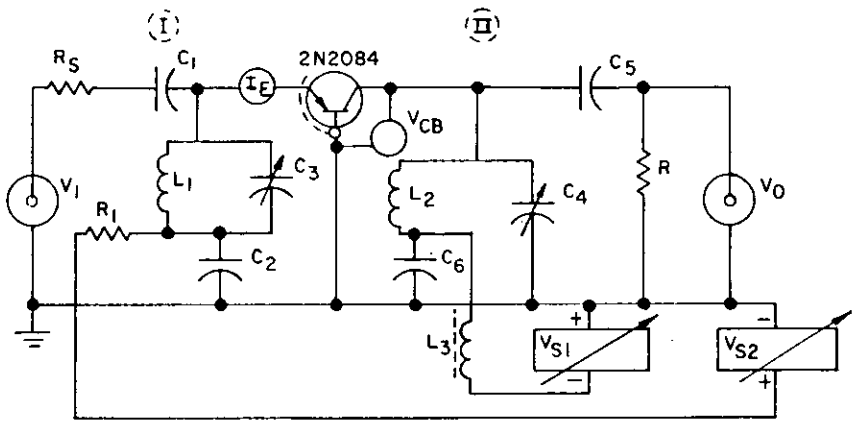
TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 6 - continued							
Forward-current transfer ratio	3076	V_{CE} = -6 Vdc; I_C = -1 mAdc		h_{FE}	32	300	---
* Subgroup 7			7				
Steady-state operation life	1027	P_T = 125 mW; V_{CB} = -12.5 Vdc; time = 340 hours (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							

* TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Resistance to solvents	---	MIL-STD-202, Method 215 (see 4.4.2)		---	---	---	---
Subgroup 2			λ= 10				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							
Subgroup 3			λ= 10				
Steady-state operation life	1026	P_T = 125 mW; V_{CB} = -12.5 Vdc (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							

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Small-signal average power gain, defined as $G_{pb} = \left(\frac{4R_S}{R_L}\right)\left(\frac{V_o}{V_1}\right)^2$

R_1=10,000 ohms

R_S=60 ohms

R is chosen such that the total impedance R_L of the tuned circuit II equals 3300 ohms

C_1 = 1,500pF

C_2 = 820pF

C_3 = 5 to 25 pF var.

C_4 = 2 to 10 pF var.

C_5 = 220pF

C_6 = 820pF

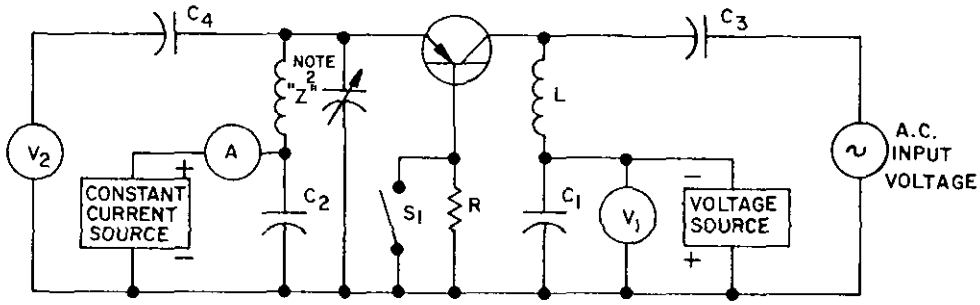
L_1 = 0.18μH

L_2 = 0.18μH

L_3=ferrite bead

FIGURE 3. Small-signal power gain circuit for type 2N2084.

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Notes:

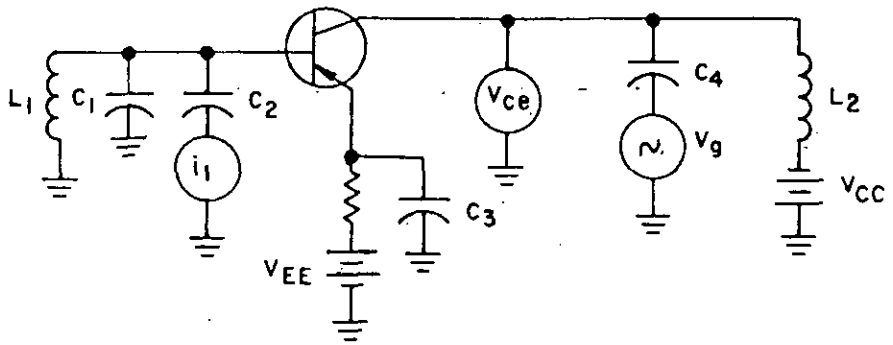
1. The constant current source is adjusted such that the specified emitter current is obtained as indicated by means of current meter A. The specified collector to base voltage is applied and measured by means of voltmeter V_1.
2. The ac impedance of the tuned circuit "Z" shall present an open circuit at the test frequency.
3. The dc resistance of L shall present a short-circuit. L is an RF choke.
4. Because of the high inner impedance between collector and base, the collector circuit effectively acts as an ac current source.
5. With switch S_1 open, a convenient reference voltage V_{ref} on voltmeter V_2 is obtained by adjusting the value of the ac input voltage.
6. With switch S_1 closed, a second reading, V_{eb} is obtained on voltmeter V_2.

   Then, $|Z_{rb}| = \frac{V_{eb}}{IC}$ where $I_C = \frac{V_{ref} - V_{eb}}{R}$

7. The capacitors C_1, C_2, C_3, and C_4, shall present short-circuits at the test frequency.

FIGURE 4. Small-signal open-circuit reverse impedance transfer ratio test circuit.

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Notes:
1. The inductance L_1 shall resonate with the capacitance C_1 and the L_1 C_1 combination shall have a large impedance compared with $1/Y_{ie}$.
2. The capacitors C_2, C_3, and C_4 shall be short-circuits at the test frequency.
3. The voltmeter V_{ce} shall be a high-impedance voltmeter and the ammeter i_1 shall be a low-impedance ammeter.

$$C_{re} = \frac{i_1}{2\pi f \quad V_{ce}}$$

FIGURE 5. Small-signal short-circuit reverse capacitance test circuit.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. *The notes specified in MIL-S-19500 are applicable to this specification.*

* 6.2 Ordering data.

(a) Lead finish if other than gold-plated (see 3.3.1).

(b) Selectivity of lead material (see 3.3.1.1).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (deletions, additions, etc.) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MU, MI
Air Force - 11, 80
DSA - ES

User activities:
Army - SM
Navy - AS, CG, MC, SH
Air Force - 13, 15, 19

Preparing activity:
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Agent:
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